

MUHAMMAD HUSSAIN ABDULLAH

MATERIALS SCIENCE ENGINEER | PEC REGISTERED (METAL/007765)

Rawalpindi, Pakistan | hussainabdullah8100@gmail.com

www.linkedin.com/in/muhammad-hussain-abdullah

SUMMARY

I am a Materials Engineer with an academic foundation and experience in materials characterization, metallography, and biomedical research. I have foundational experience in non-destructive testing techniques, failure analysis, inspection procedures, degradation mechanisms, and materials analysis techniques. Through academic research projects and practical lab work, I have developed strong problem-solving skills, adaptability, and passion for continuous learning in dynamic environments.

EDUCATION

BS in Materials Science & Engineering

Nov 2021 - Aug 2025

Institute of Space Technology (IST), Islamabad, Pakistan

PROFESSIONAL EXPERIENCE

Engineer Intern - Scientific and Engineering Service Directorate

Jul 2024 - Aug 2024

- Performed Non-Destructive Testing on metallic components to support inspection and integrity assessment in accordance with ISO standards.
- Assisted in evaluation of material degradation, surface defects, and fabrication-related flaws in welded joints and components.
- Supported failure investigations and root-cause analysis through metallographic examination and test result interpretation.
- Prepared technical inspection reports and documentation for quality assurance and integrity verification.

Research Assistant - Bioengineering Research and Development Pvt Ltd

Jun 2023 - Jul 2025

- Developed hydrogels for tissue engineering through novel process design and materials optimization.
- Performed materials characterization (TGA, FTIR), antibacterial assays, CAM assays, and degradation assessments to evaluate material performance.
- Conducted failure analysis of steel and brass components, identifying defects, microstructural anomalies, and potential degradation mechanisms.
- Supported technical documentation, data analysis, and reporting, ensuring traceability and compliance with engineering protocols.

TRAINING & PROFESSIONAL DEVELOPMENT

- | | |
|---|--------------|
| Participation in "Metallurgical Failure Analysis in Practice" (1 CPD Point) | 18 Sept 2025 |
| Participation in "Corrosion and Materials Protection" (1 CPD Point) | 25 Nov 2025 |
| Participation in "Procurement & Supply Chain for Projects" (1 CPD Point) | 22 Feb 2026 |

PUBLICATION

Haseeb Nawaz, Muhammad, et al. "Advancing Cartilage Repair: A Biomimetic Approach with Oxidized Guar Gum, Chitosan, Polyether Ether Ketone-Based Injectable Hydrogel." *Materials Letters*, vol. 371, Sep. 2024, p. 136950. *ScienceDirect*, <https://doi.org/10.1016/j.matlet.2024.136950>.

TECHNICAL SKILLS

Failure Analysis

Software: Origin(Data Analysis), Ansys (Mechanical)

Microsoft Office Suite

Material Characterizations (TGA, FTIR)

NDT Techniques

Metallography

ADDITIONAL INFORMATION

- Languages:** Urdu, English
- Interests:** Computational Materials Science, Biomaterials, Corrosion